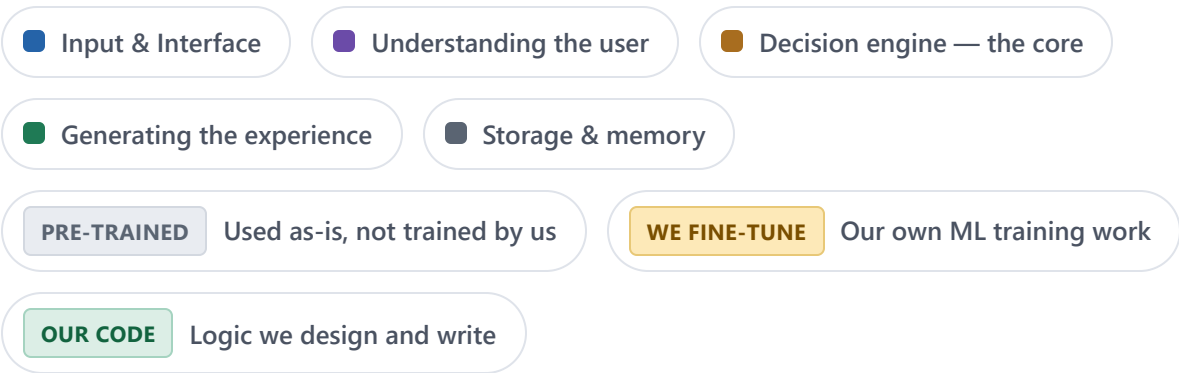


Personalized AI-Based Narrative Content Creation and Voice Generation Using Generative AI

A person types or speaks a worry. Our system understands the emotion behind it and the negative thinking pattern inside it, selects a scientifically evidence-backed psychological technique that applies to that specific pattern, and generates a personalized narrative — delivered as text, image, and spoken voice, and saved so the user can revisit it.

A How to read this diagram

Every stage is colour-coded by what it does, and tagged by who built it. The tags matter in your review — they show exactly where your own machine-learning work sits.



B The complete workflow, stage by stage

Read top to bottom. This is exactly what happens between the user pressing "Send" and seeing their result.

PHASE 1 — THE USER SPEAKS

Getting the thought into the system as plain text

1

User gives their thought

HTML · CSS · JavaScript

A simple web page with a text box and a microphone button. The user either types their worry or records it in their own voice.

Output → typed text, or a short audio recording (max 30 seconds)



Input travels to the backend

Flask

OUR CODE

2

Flask is the receptionist. It receives the request from the browser, passes it into our pipeline, and later carries the result back.



Speech becomes text

Whisper

PRE-TRAINED

3

Only runs if the user spoke. It writes down what was said. If the user typed instead, this stage is skipped completely.

Key design point → voice is just an optional front door. After this stage **every path is identical** – the rest of the system only ever handles text. There is no second pipeline.



Safety check — runs before anything else

OUR CODE

!

Scans for signs of genuine crisis or self-harm. If detected, the system **stops here**: it generates nothing and instead shows a calm message with real helpline information (Tele-MANAS 14416 · KIRAN 1800-599-0019).

Why this is on the diagram → our tool takes emotional input from real people. It is a self-reflection tool, not therapy. This gate is the first thing that runs, always.



PHASE 2 — THE SYSTEM UNDERSTANDS

Turning a sentence into structured knowledge about the person

Language analysis + thinking-pattern detection

spaCy

OUR RULES

4

spaCy breaks the sentence into its grammatical parts — subject, verb, negation, named things. Our own rules then look for **cognitive distortions**: the specific ways people distort their own thinking.

Examples → "always / never" → all-or-nothing thinking · "I'll definitely fail" → predicting the worst · "I'm not good enough" → self-labeling



5

Emotion detection

DistilBERT

WE FINE-TUNE THIS

Reads the same text and predicts which emotions are present, and how strongly. **This is one of our two real machine-learning training tasks** — we fine-tune it ourselves on the GoEmotions dataset.

Output → fear 0.61 • sadness 0.22 • anger 0.04 ...



6

Context Engine builds the "Case Frame"

OUR CODE

Plain Python, no AI. It packs everything learned so far into one neat structured object: the original words, the emotions, the thinking patterns, the underlying goal, and a one-line summary of the real concern.

Why this matters → every later stage reads only this one package. That is what lets four team members build four modules separately and still have them fit together.

**PHASE 3 — THE SYSTEM DECIDES**

This is the heart of the project — the part that makes it more than a chatbot

7

Scientific Principle Selector

OUR KNOWLEDGE BASE + LOGIC

We build a library of about 15 **concept cards**. Each card holds one evidence-backed psychological technique — what it is, the research paper it comes from, how strong that evidence is, when it applies, how to apply it, and its limitations.

The system matches the Case Frame against these cards and selects the techniques that genuinely fit this person's situation.

Example selection → Cognitive reframing (primary) • Self-efficacy (supporting) • Implementation intentions (concrete action)

Say this in your review: without this stage we would be a motivational-quote generator. With it, **every single output can be traced back to a named psychological principle and a real research citation.** Nothing the system says is invented on a whim.

**PHASE 4 — THE SYSTEM CREATES**

One thought becomes three kinds of experience

Personalized narrative generation

FLAN-T5 + LoRA

WE FINE-TUNE THIS

Our second real training task. The chosen technique supplies the **structure**; the model supplies the **personalization**, rewriting each part using the user's own words and situation. It always follows four beats:

1. **Current reality** – acknowledge the worry honestly, never dismiss it
2. **Reframe** – a more accurate, balanced way of seeing it
3. **Desired future** – the outcome the person actually wants
4. **Concrete next step** – one small, specific, doable action

Strict rule → the system never says *"Don't worry, you'll definitely succeed."* Empty promises are not reframing. It gives a realistic, grounded, actionable perspective.



Narrative text shown to the user immediately

OUR CODE

We do not make the user wait for the image and audio. The text appears first, then voice, then the image. The experience feels fast even while work continues in the background.



Voice narration

Kokoro-82M

PRE-TRAINED

Reads the narrative aloud in a calm, natural human-sounding voice. Because the whole point of this project is emotional comfort, a robotic voice would undermine the experience.



Visual generation

Stable Diffusion (Turbo)

PRE-TRAINED

A short description is taken from the "desired future" beat and turned into an image, generated in the background so it never blocks the user. A pre-generated fallback image is used if live generation is slow or fails.



PHASE 5 — THE SYSTEM REMEMBERS

What turns a one-off tool into a personal companion

Everything is saved

SQLite

OUR SCHEMA

The original input, the emotions, the thinking patterns, the technique chosen, the narrative, the image, and the audio are all stored together as one complete session.



The user revisits, and the system personalizes

[OUR CODE](#)

A "My Journey" page lets the user reopen any past narrative, image, and voice. The system also notices recurring themes — *"you have raised career confidence three times"* — and feeds that back into future narratives.

This closes the loop → the same worry typed by two different people produces two different narratives. The same worry typed twice by **one** person produces a narrative that remembers.

C The same journey, with a real example

Follow one sentence all the way through the system. This is the single most convincing thing you can walk your guide through.

① User says	"I am scared I will fail my placement interviews. I don't think I'm good enough."
④ Thinking patterns found	<code>fortune_telling</code> <code>self_labeling</code> — predicting a bad outcome as if certain, and attaching a permanent label to oneself
⑤ Emotions detected	<code>fear 0.61</code> <code>self-doubt 0.48</code> <code>sadness 0.22</code>
⑥ Case Frame	Goal: succeed in placement interviews, feel confident Summary: <i>fear of interview failure driven by low self-belief</i>
⑦ Techniques selected	Cognitive reframing — because a distorted thought was detected Self-efficacy building — because self-doubt is present Implementation intention — because the user needs a concrete next action
⑧ Narrative generated	CURRENT REALITY You are facing interviews that matter to you, and the fear you feel is real. REFRAME "I'm not good enough" is a feeling, not an assessment. Nervousness before something important is your body preparing — not a prediction of the result. DESIRED FUTURE You walk into the room having already answered these questions many times, and your preparation speaks before your nerves do. CONCRETE NEXT STEP Tomorrow evening after dinner, you will do one 20-minute mock interview with a friend.
⑩ ⑪ Delivered as	Text on screen · calm voice narration · an image of the confident interview scene
⑫ ⑬ Saved	Stored as one session. Next time this user mentions career confidence, the system knows.

D The same system, seen as layers

If your guide asks "show me the architecture," this is the view they expect.

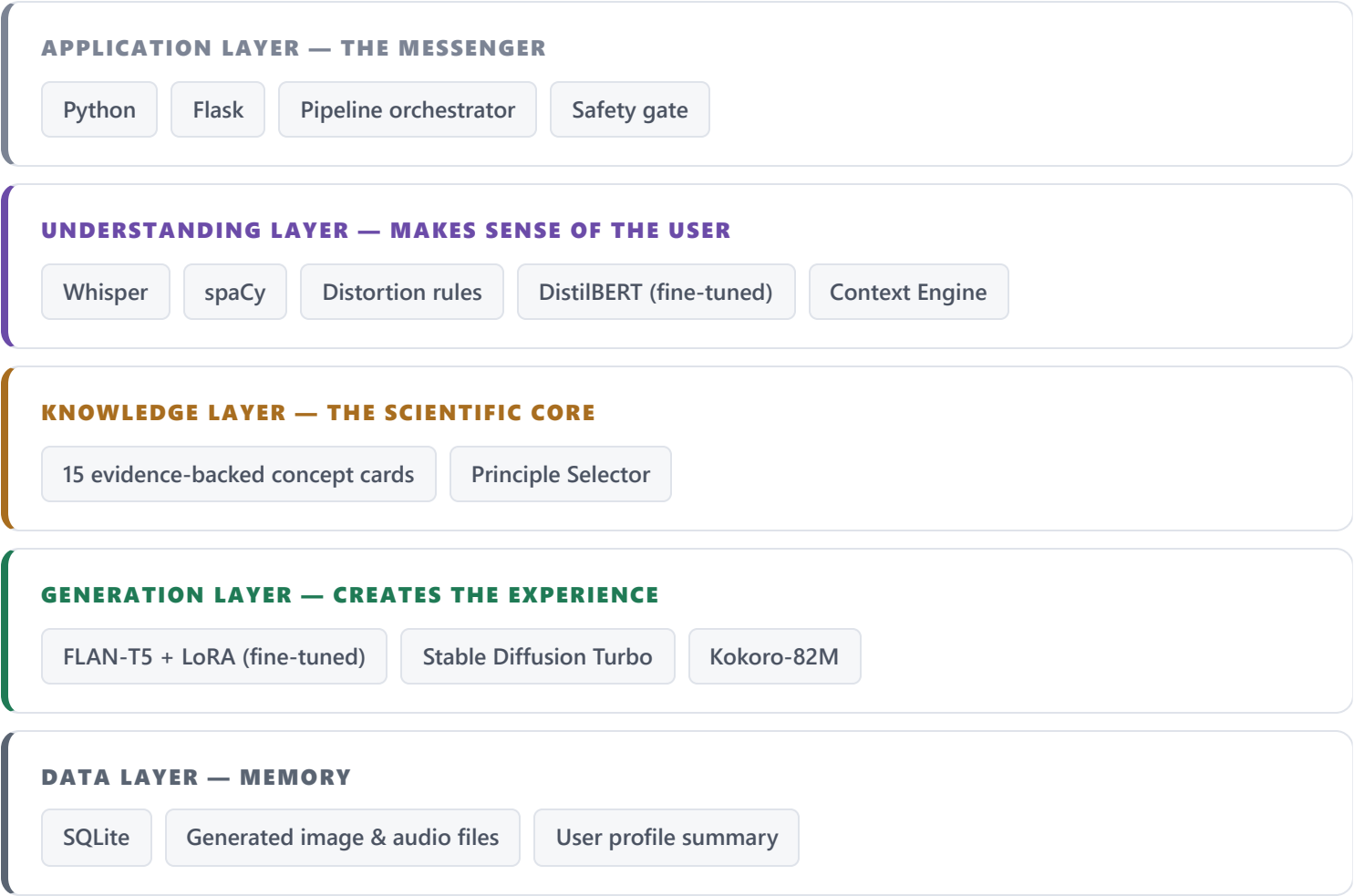
PRESENTATION LAYER — WHAT THE USER SEES

HTML

CSS

JavaScript

Browser audio recording



E What is borrowed, and what is genuinely ours

Expect the question "*did you actually build AI, or just call libraries?*" — this table is your answer.

Pre-trained, used as-is

Standard, already-solved problems. Rebuilding these would be neither realistic nor valuable.

- **Whisper** — speech recognition
- **spaCy base model** — grammar parsing
- **Stable Diffusion** — image generation
- **Kokoro-82M** — voice synthesis

Our own contribution

This is where the machine learning and the system design actually live.

- **DistilBERT fine-tuning** — trained by us on a mapped GoEmotions dataset, evaluated with macro-F1
- **FLAN-T5 LoRA fine-tuning** — trained on a dataset of ~500 examples we construct ourselves
- **The scientific knowledge base** — 15 concept cards, each with a real research citation and an honest evidence rating
- **Context Engine & Principle Selector** — the logic that makes output grounded instead of generic
- **Cognitive-distortion rules** — hand-authored and measured for precision and recall
- **The safety system** and the full end-to-end integration

F How to present this in 90 seconds

Walk your guide down the diagram while saying roughly this.

SUGGESTED SPEAKING SCRIPT

"Sir/Ma'am, our system takes a person's worry and turns it into a personalized, scientifically grounded narrative delivered as text, image and voice.

It begins here — the user types or speaks. If they speak, Whisper converts it to text, so from that point onward everything is one single pipeline. Before anything else runs, a safety check looks for signs of genuine crisis and, if found, shows helpline information instead of generating content.

Then the system understands. spaCy plus our own rules detect the negative thinking pattern — here, predicting the worst and self-labeling. A DistilBERT model that **we fine-tuned ourselves** detects the emotions. Both results are packed into one structured object we call the Case Frame.

This next stage is the heart of our project. We built a library of evidence-backed psychological techniques — cognitive reframing, self-efficacy, implementation intentions — each one with the actual research paper it comes from and an honest note on its limitations. The system matches the user's situation against this library and selects the technique that genuinely applies. This is why we are not a motivational-quote generator: every output traces back to a named principle and a real citation.

Then it creates. A FLAN-T5 model that **we also fine-tuned** writes a four-part narrative — current reality, reframe, desired future, and one concrete next step. It never makes empty promises like "you'll definitely succeed." The text appears immediately; Kokoro narrates it in a natural voice; Stable Diffusion generates a matching image in the background.

Finally it remembers. Everything is saved to SQLite, the user can revisit any past session, and recurring themes make future narratives more personal.

To be clear about our own contribution — we use pre-trained models for the solved problems like speech and image generation. Our machine-learning work is the two fine-tuned models, and our engineering contribution is the knowledge base and the decision engine that grounds every output in evidence. The whole system runs offline on a normal laptop, with no paid API."